

ADC project report

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Problem-statement:

In many modern environments, wireless communication suffers from **network congestion, security threats, and electromagnetic interference**. Additionally, in critical areas like hospitals and aircraft, RF-based communication can be unsafe or restricted.

This project proposes a **Li-Fi-based communication system** that uses visible light for data transmission, offering a **secure, interference-free, and efficient alternative** for next-generation wireless communication.

Aim:

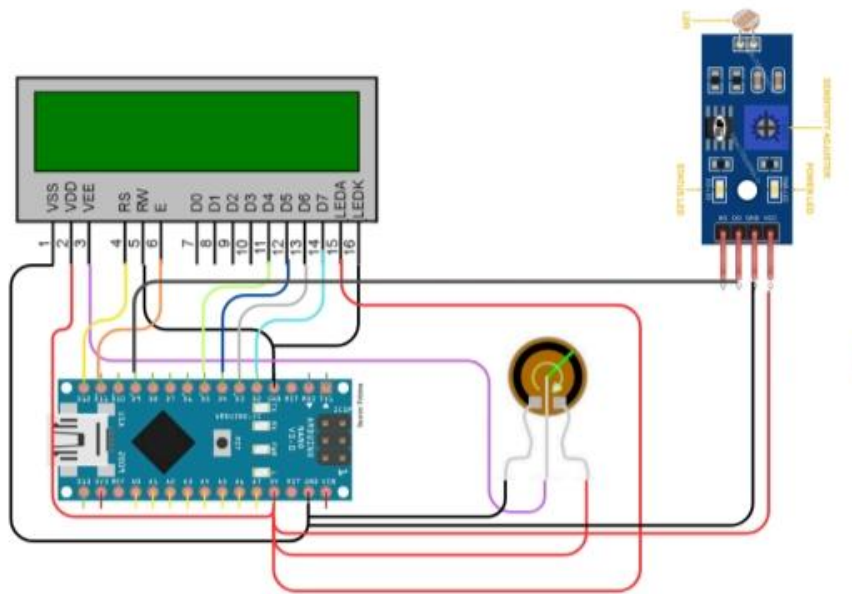
To design and implement a **Li-Fi based wireless communication system using Arduino**, capable of transmitting data through light signals and displaying the received information on an LCD, for use in secure and interference-free real-world environments.

Objectives:

1. To understand the working principle of **Li-Fi communication technology**.
2. To build a transmitter and receiver circuit using Arduino, LED/IR sensor, and resistors.
3. To transmit data wirelessly using light signals instead of radio waves.
4. To receive and decode the transmitted data using an IR sensor.
5. To display the received data on an LCD screen.
6. To analyze the performance and limitations of the system.

7. To study real-world applications such as hospitals, aircraft, and smart systems.
8. To develop practical skills in Arduino programming and hardware interfacing.

Block Diagram:



Explanation

This setup represents the **receiver section** of a Li-Fi communication system using an Arduino.

1. Arduino NaNO (Main Controller)

- The **Arduino NaNO** acts as the brain of the system.
- It receives signals from the sensor, processes them, and sends output to the LCD display.
- Power is supplied through the 5V and GND pins.

2. IR / Light Sensor Module

- This module detects **light signals (Li-Fi data)** transmitted from the transmitter side (LED).

- It converts the received light into **electrical signals**.
- Pins:
 - **VCC** → Connected to 5V
 - **GND** → Ground
 - **DO (Digital Output)** → Connected to Arduino input pin

When light is received, the sensor sends HIGH/LOW signals to Arduino.

3. LCD Display (16x2)

- The LCD is used to **display the received message/data**.
- It is connected to Arduino using multiple digital pins.
- A resistor is used to control **contrast or current**.

Example: If light signal represents text, it will be shown on the LCD.

4. Resistor

- Used to **limit current** and protect components.
- Also helps in proper functioning of LCD contrast or signal stability.

5. Breadboard

- Used for **easy connections** without soldering.
- Helps organize the circuit neatly.

6. Power Supply Connections

- **Red wires** → 5V supply
- **Black wires** → Ground (GND)
- These lines distribute power to all components.

Result

The **Li-Fi based wireless communication system using Arduino** was successfully designed and implemented. The system was able to **transmit data through light signals** and receive it accurately using the IR sensor module.

The received data was **successfully displayed on the LCD screen** in real time, demonstrating effective communication between the transmitter and receiver units.

The project proved that:

- Data can be transmitted wirelessly using **light instead of radio waves**.
- The system works efficiently over a short distance with **minimal delay**.
- Communication is **secure and free from electromagnetic interference**.

However, it was observed that:

- The system requires a **direct line of sight** between transmitter and receiver.
- Performance is affected by **obstacles or strong ambient light**.

Conclusion:

The project demonstrates that Li-Fi technology is a **promising alternative to traditional wireless communication**, especially in environments where RF communication is limited or unsafe.